

► VISION PROCESSING PIPELINE

INPUT IMAGE I

$3 \times 224 \times 224$ RGB Image
 14×14 Grid of 16×16 Visual Patches

$$\mathbb{R}^{B \times 3 \times 224 \times 224}$$

FROZEN VISION BACKBONE

Pretrained ViT-B/16 Encoder (85.8M)
 196 Patch Embeddings $\times$ 768 -dim

$$\mathbb{R}^{B \times 196 \times 768}$$

VISION PROJECTION HEAD

Linear Projection Layer
Dimension Reduction: $768 \rightarrow 128$

$$\mathbb{R}^{B \times 196 \times 128}$$

STRUCTURE-AWARE HEDO ADAPTER

Hamiltonian Dissipative Dynamics
Learned Skew-Symmetric Coupling & Damping

$$\mathbb{R}^{B \times 196 \times 128}$$

NATIVE MAMBA-2 SEQUENCE BLOCK

State Space Duality (SSD)
Chunkwise Linear $O(L)$ Sequence Recurrence

$$\mathbb{R}^{B \times 196 \times 128}$$

SPATIAL MEAN POOLING HEAD

Global Mean Patch Aggregation
 ℓ_2 -Normalized Representation ($\mathbf{z}_v \in \mathbb{R}^{128}$)

$$\mathbf{z}_v \in \mathbb{R}^{128}$$

► LANGUAGE PROCESSING PIPELINE

INPUT CAPTION T

Natural Language Text ($L \leq 64$ Tokens)
Tokenized Wordpiece Sequence

$$\mathbb{Z}^{B \times L}$$

FROZEN LANGUAGE BACKBONE

Pretrained RoBERTa-base (124.6M)
 L Wordpiece Embeddings $\times$ 768 -dim

$$\mathbb{R}^{B \times L \times 768}$$

LANGUAGE PROJECTION HEAD

Linear Projection Layer
Dimension Reduction: $768 \rightarrow 128$

$$\mathbb{R}^{B \times L \times 128}$$

STRUCTURE-AWARE HEDO ADAPTER

Hamiltonian Dissipative Dynamics
Learned Skew-Symmetric Coupling & Damping

$$\mathbb{R}^{B \times L \times 128}$$

NATIVE MAMBA-2 SEQUENCE BLOCK

State Space Duality (SSD)
Chunkwise Linear $O(L)$ Sequence Recurrence

$$\mathbb{R}^{B \times L \times 128}$$

MASKED SEQUENCE POOLING HEAD

Attention-Masked Mean Aggregation
 ℓ_2 -Normalized Representation ($\mathbf{z}_t \in \mathbb{R}^{128}$)

$$\mathbf{z}_t \in \mathbb{R}^{128}$$

SYMMETRIC MULTI-POSITIVE INFONCE ALIGNMENT HEAD

Grouped Sampling: 8 Unique Images $\times$ 5 Captions = 40 Cross-Modal Pairs per Batch

Similarity Metric: Temperature-Scaled Cosine Similarity with Learned Temperature τ

Alignment Objective: Bidirectional Symmetric Loss ($\mathcal{L}_{i2t} + \mathcal{L}_{t2i}$) with Multi-Positive Supervision